

HALT: HELD-OUT-SOURCE EVALUATION FOR HALLUCINATION DETECTION IN FORENSIC VIDEO REPORTING

Anonymous authors

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ABSTRACT

Hallucination detectors for long-form multimodal generation are almost always evaluated on text from the same source LLMs they were trained on. In forensic video reporting we show this conflates detection with identifying which LLM wrote the text. A detector that silently stops working when the underlying model is updated offers false assurance, in a setting where inventing a detail and omitting a crime are two different evidentiary failures. We build HALT (Hallucination Across LLMs and Techniques) on a previously released corpus of 19,361 forensic video reports (21.8M words) in which three frontier multimodal LLMs each describe all 807 surveillance videos, spanning eleven crime categories, under all eight prompting strategies, labelled for six hallucination types by a cross-judging panel in which no model evaluates its own output. Our central result is that much of what a detector appears to know is the identity of the model that wrote the report: a predictor reading no report text at all, returning only the writing model’s base rate, captures 70 to 73% of a detector’s above-chance signal on the fabrication labels against 13% on omission. Withholding a source LLM accordingly costs the fabrication labels 0.104 to 0.191 AUC against 0.035 to 0.105 for omission and distortion, consistently across four detector families from bag-of-words to a fine-tuned long-context encoder. Two further findings concern the panel’s H1–H6 labels: (i) report length predicts the panel’s scene-fabrication label at AUC 0.727 but the consensus of two human annotators who each independently labelled the same 130 reports at only 0.566, and (ii) models are not allowed to judge their own output, so each source LLM is scored by a different pair of judges; those pairs differ in how often they agree, so comparing hallucination rates across models partly compares their judges. We release the reliability tiers, the four splits, and reference implementations of every detector.

1 INTRODUCTION

Large language models are increasingly evaluated by other large language models. Panel-of-judges protocols now label datasets far larger than human annotation could reach, and those labels underpin leaderboards, safety audits and deployment decisions. Whether they measure what they are taken to measure is rarely examined, and the raw verdicts are rarely released.

Hallucination detection inherits both problems. A detector is trained on text labelled this way and tested on text from the same LLMs that produced the training set, split at random. Model families differ sharply in surface style and equally sharply in how often they hallucinate, so a detector with access to both regularities can score well by recognising which family wrote a passage and applying its base rate, without assessing whether any claim is supported. Under a random split this is invisible: the models at test time are the models at training time.

The distinction matters wherever a detector meets text from a model it has not seen, which is the normal case. We study it in forensic video reporting and scope our claims to that domain. There, inventing a detail and omitting a crime are different evidentiary failures, a screening tool that silently stops working when the underlying model is updated offers false assurance, and expert reference

annotations written independently of any model are available, which most long-form generation tasks lack.

Separating the two possibilities requires text from several model families over identical inputs, labelled per instance, so the source LLM can be withheld while everything else is fixed. We therefore ask:

1. Do hallucination detectors transfer to source LLMs they were not trained on, and by how much does performance change?
2. How does detectability depend on hallucination type, and does the type ordering hold under transfer?
3. Which surface properties carry the detectable signal, and what remains once source identity is removed?
4. How much of the loss does conditioning on the reference annotation recover, and for which types?
5. How does panel design shape the labels themselves, both in what drives judge verdicts and in what excluding self-evaluation implies?

The report corpus, the labelling protocol, the human-validation subset and the per-judge verdicts are inherited from prior work (Anonymous, 2026), which measured how often hallucination occurs and released its materials. Its detectability analysis trained and tested every detector on the same three source LLMs, so it could not ask whether detection transfers. New here are: the held-out-source split and the three other video-disjoint partitions that make the transfer question askable; reference-conditioned detectors and fine-tuned long-context encoders, so the result cannot be attributed to weak models; per-type reliability tiers; a length-bias diagnosis scoring the panel and human labels against the same feature; and a within-judge test showing that excluding self-evaluation confounds the source LLM with the judges who scored it.

Throughout, we refer to the LLM that produced a report as its *source LLM*, to distinguish it from the LLMs that later judge it. In the language of shortcut learning (Geirhos et al., 2020), source identity is a spurious feature: predictive in distribution, cheaply recoverable from surface style, and worthless once the source changes.

2 RELATED WORK

Existing hallucination benchmarks target short outputs and a fixed set of source LLMs. CHAIR (Rohrbach et al., 2018) measures object hallucination in captions, POPE (Li et al., 2023b) uses polling, HaluEval (Li et al., 2023a) benchmarks 35K samples, THRONE (Kaul et al., 2024) unifies evaluation across six models, Bingo (Cui et al., 2023) identifies bias triggers, and VideoHalluciner (Wang et al., 2024c) extends the setting to short clips. None varies the writing model while holding the input fixed, so none can separate a detector that reads evidence from one that reads style. Detection methods inherit the same limitation: SelfCheckGPT (Manakul et al., 2023), semantic entropy (Kuhn et al., 2023), and FActScore (Min et al., 2023) detect hallucination via consistency, uncertainty, or fact decomposition, but are evaluated within a single source LLM or across several pooled without a held-out condition. Our reference-conditioned baseline is closest to FActScore, simplified to lexical grounding. The closest analogue to our argument lies outside the hallucination literature: Wang et al. (2024b) show that detectors of machine-generated text degrade on generators they have not seen. That work asks whether text came from a machine; we ask whether a detector nominally judging content is keyed to the same signal.

Panel designs (Zheng et al., 2023) and cross-model panels that mitigate self-preference (Panickssery et al., 2024) are now standard, and their reliability is typically summarised by a single agreement figure. A growing body of work questions whether that figure means what it appears to: Wang et al. (2024a) document position and scoring biases, Wu & Aji (2023) show judges reward stylistic properties over substance, and Stureborg et al. (2024) find judges inconsistent across prompts and biased toward text they find predictable. Length bias sits in this family. We report reliability per hallucination type against independent human raters, measure length bias against human labels on the same reports (Section 4.6), and show that the no-self-evaluation constraint itself confounds each

source LLM with its judges, which single-figure reporting hides. Section 3.1 states what is inherited from prior work and what is new.

3 METHODOLOGY

3.1 TASK AND CORPUS

An instance is a triple (v, m, τ) : a surveillance video v , a source LLM m , and a prompting strategy τ . The source LLM produces a free-form forensic report r describing what occurs in v , and each report carries an expert reference annotation g_v written independently of any model. The task is to predict, from r and optionally g_v , six binary labels: H1 scene fabrication, H5 entity fabrication and H6 phantom actors, which invent content absent from the footage; H3 crime missed, which drops the criminal act; and H2 crime misclassification and H4 severity minimization, which misstate its nature or gravity. These group onto three axes, *fabrication*, *omission* and *distortion*. We report per-type AUC and F_1 , a macro over the six, a macro restricted to the reliable types (Section 3.2), and a binary any-hallucination score.

Provenance. The reports, taxonomy, panel protocol, per-judge verdicts and human-validation subset were introduced and released by prior work (Anonymous, 2026) to measure hallucination prevalence, and HALT reuses all of it as raw material. New here are the held-out-source formulation of the detection task, the reliability tiering, the four splits, the reference-conditioned and encoder detectors, and every analysis in Section 4.

The inherited corpus is 19,361 reports on 807 UCF-Crime videos (Sultani et al., 2018) with UCA annotations (Yuan et al., 2024) as reference, written by three frontier multimodal LLMs under eight prompting strategies. Two of its properties make the held-out-source question askable. It is fully crossed up to seven failed API calls, so withholding a source LLM holds the input fixed by construction. And verbosity varies by a factor of three under an identical task, 580 mean words per report for GPT against 1,802 for Gemini, the regularity Section 4 shows detectors exploiting.

3.2 LABELS AND THEIR RELIABILITY

The inherited labels come from a panel of the same three models acting as judges (Zheng et al., 2023), each report scored by the two that did not write it, so self-preference bias (Panickssery et al., 2024) is structurally excluded (Anonymous, 2026). Three properties of that design matter here. It yields two verdicts per report under a conjunctive rule, so alternative rules can be applied post hoc (Section 4.1). Judge inputs cap the reference at 1,500 and the report at 3,000 characters, which Section 4.6 shows shapes the labels. And no model judging its own output means each source LLM is scored by a distinct judge pair, the confound of Section 4.6.

An inherited subset of 130 reports carries independent labels from two human annotators, members of the research group rather than forensic practitioners, who worked from text with the panel’s labels hidden. From the rater-rater and panel-consensus agreements we assign each type a reliability tier and exclude the low tier from the headline macro; the tiering is ours, and Section 4.1 reports it.

3.3 EVALUATION SETUP

Each video contributes 24 reports, so a split over reports would let a detector memorise the video rather than learn the phenomenon; Figure 1 shows the three partitions. The standard and held-out-strategy splits are therefore video-disjoint. The held-out-source split is not, since every video is described by every source LLM; Section 4.3 reports a disjointness-enforced variant under which the effect is larger, so the main tables are conservative.

We provide reference detectors in two families. The first is linear: five logistic regressions with balanced class weights, differing only in features. **Surface** uses verbosity, sentence statistics, hedging density, numeric detail density, type-token ratio, truncation length, and source and strategy identity indicators; **surface-noid** drops the identity features; **tfidf** uses word 1–2 gram TF-IDF; **embed** uses sentence embeddings; and **grounded** uses report-versus-reference features only, namely the rate of report tokens absent from the reference, the rate of reference tokens absent from the report,

Jaccard overlap, reference coverage, length ratio and TF-IDF cosine. Identity features are dropped automatically on any split where they are constant in training and unseen at test.

The second family is fine-tuned encoders. One encoder per split, six per-type heads plus a binary head, report and reference as a sentence pair, positive weights matching the linear baselines. Context length is chosen rather than defaulted: the judge saw about 1,125 tokens, so a detector at 512 sees less than the judge did. We report **DeBERTa-v3-base** at its 512-token limit, **ModernBERT-base** at 1,280, matching the judge’s view, and ModernBERT at 2,048, exceeding it.

The corpus, per-judge verdicts, aggregated labels and rubrics were released with the prior study. HALT adds, and releases: the reliability tiers with supporting n , the four splits, reference implementations of every detector, and a datasheet (Gebu et al., 2021). The artifact is available at <https://anonymous.4open.science/r/HALT-ANON-7FF2/>. Supporting tables cited below are in Appendices A to H of this paper.

4 RESULTS

4.1 CORPUS STATISTICS AND LABEL PROPERTIES

The prior study reports 91.1% of reports carrying at least one hallucination under the released rule. What it does not report is how far that depends on the rule: a disjunctive rule on the same verdicts gives 98.5% and 3.55 per report, a seven-point swing that a release of aggregated labels alone would conceal.

Reliability is strongly type-dependent (Table 1): entity fabrication, misclassification and phantom actors reach κ of 0.81, 0.70 and 0.66 against the human consensus, scene fabrication only 0.15, and we exclude it from the headline macro. Two qualifications apply. The consensus κ is computed

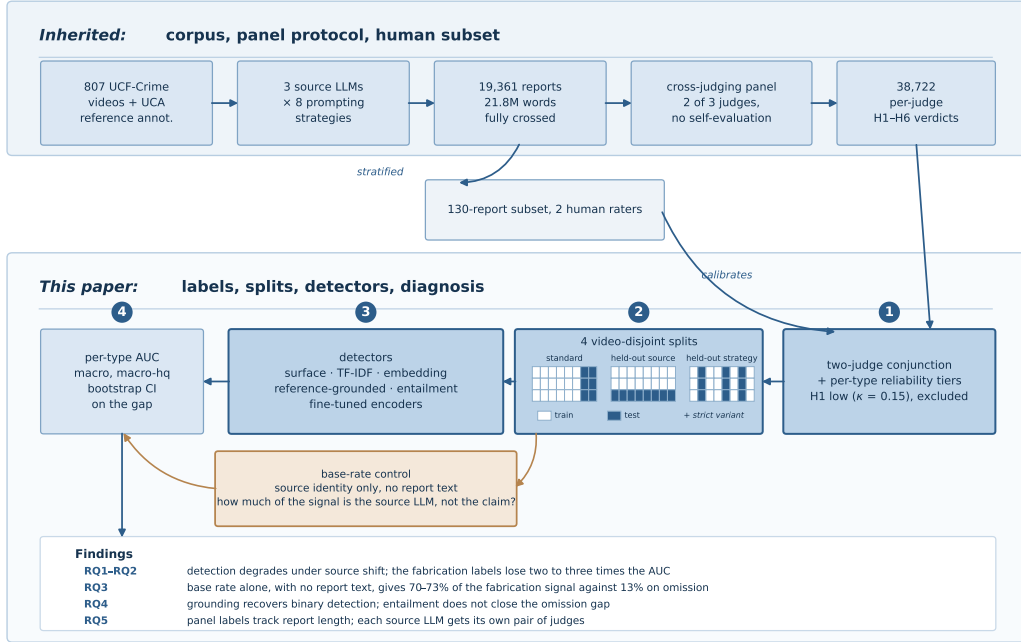


Figure 1: The HALT pipeline. The shaded band is inherited; the remainder is this paper: aggregating the raw verdicts into labels whose per-type reliability is calibrated against an independently scored subset, partitioning the corpus into four video-disjoint splits, and evaluating four detector families against a base-rate control that reads no report text. In step 2 each cell is one report, rows are source LLMs and columns are video-by-strategy conditions; existing benchmarks admit only the standard split.

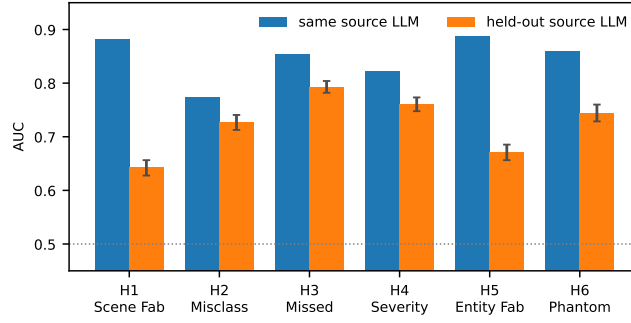


Figure 2: Per-type detector AUC (TF-IDF features) under the standard same-source split and under leave-one-source-out, with bootstrap 95% intervals on the held-out bars. The fabrication types (H1, H5) lose roughly four times as much as the omission and distortion types, inverting the ordering a same-source evaluation would report.

only where the two raters agree, a subset of 62 to 110 of 130, so H5’s 0.81 rests on 64 reports and, since the restriction selects the straightforward cases, κ_{panel} is an upper bound. The raters also agree only modestly with each other on fabrication (macro $\kappa = 0.28$, $\kappa \approx 0$ on H1), and the disagreement has a direction: because references are summary-level, granular detail absent from the summary can be scored as fabrication. Absolute fabrication prevalence is therefore an upper bound; relative comparisons are unaffected, since the bias is shared across source LLMs. Because judge inputs truncate the report at 3,000 characters and the most verbose source LLMs are the most truncated, truncation can only understate their fabrication rates.

4.2 SAME-SOURCE DETECTION REPRODUCES PRIOR FINDINGS

Under the standard video-disjoint random split we reproduce the prior study’s detectability finding, including that source identity is its most informative single feature: surface features reach 0.858 on scene fabrication against 0.646 on crime omission, and TF-IDF reaches 0.846 macro and 0.820 binary. That is what a practitioner would conclude under conventional evaluation.

4.3 DETECTION OF THE FABRICATION LABELS DEGRADES MOST (RQ1–RQ2)

Every detector degrades under leave-one-source-out, and every one degrades unevenly (Table 2, Figure 2): fabrication falls by 0.104 to 0.191 AUC against 0.035 to 0.105 for omission and distortion, a ratio of 1.4 to 3.1. We describe this as degradation of the fabrication *labels*: the raters agree least on exactly these types (Section 4.1), so a property of fabrication and a property of how fabrication is labelled here cannot be separated. For the lexical detector the loss inverts the per-type ordering outright, TF-IDF giving fabrication 0.686 against omission 0.793 under shift having given 0.876 against 0.854 in distribution.

Table 1: Per-type label reliability on the 130-report human-annotated subset. κ_{panel} compares the panel against the human consensus and is computed only on reports where the two raters agree; n is that subset size. Tier thresholds are $\kappa \geq 0.65$ (high) and $\kappa \geq 0.40$ (medium). H1 is excluded from the headline macro score.

Type	Axis	n	$\kappa_{\text{human-human}}$	κ_{panel}	Tier
H1 Scene Fabrication	Fabrication	62	−0.008	0.150	low
H2 Crime Misclassification	Distortion	99	0.470	0.700	high
H3 Crime Missed	Omission	110	0.532	0.432	medium
H4 Severity Minimization	Distortion	103	0.406	0.583	medium
H5 Entity Fabrication	Fabrication	64	0.082	0.811	high
H6 Phantom Actors	Fabrication	84	0.192	0.658	high
Macro			0.279	0.556	

Table 2: Degradation from the standard split to leave-one-source-out, by axis, for four detector families. Drops are LOSO minus standard. The ratio is the fabrication drop divided by the larger of the omission and distortion drops, the conservative choice; it is quoted in this form throughout. Fabrication degrades most in every family.

Detector	drop by axis			MACRO	ANY		
	fabr.	omis.	dist.		standard	LOSO	ratio
tfidf (linear)	−0.191	−0.061	−0.054	−0.123	0.820	0.566	3.1×
DeBERTa-v3, 512	−0.104	−0.035	−0.043	−0.072	0.861	0.729	2.4×
ModernBERT, 1,280	−0.163	−0.089	−0.097	−0.129	0.833	0.644	1.7×
ModernBERT, 2,048	−0.145	−0.079	−0.105	−0.120	0.832	0.642	1.4×

The difference is statistically unambiguous. Bootstrapping the fabrication degradation minus the omission-and-distortion degradation over 1,000 resamples of both splits gives -0.134 with a 95% interval of $[-0.148, -0.121]$, excluding zero. Per type (Table 7): bootstrap 95% intervals place crime omission above 0.776 in every fold, while scene fabrication never exceeds 0.714 and entity fabrication never 0.730, and the intervals do not overlap in any fold.

The effect is not an artifact of residual video overlap: enforcing video disjointness as well lowers every number and widens the gap, fabrication to 0.668 against omission 0.780 and binary detection to 0.559 (Appendix D).

For the lexical detector binary detection all but disappears: any-hallucination AUC falls from 0.820 to 0.566, and on the held-out Claude fold the interval is $[0.502, 0.578]$, indistinguishable from a coin flip.

Nor is it an artifact of the low-reliability type. Dropping H1 from the fabrication axis, as the headline macro already does, leaves the ordering intact: the ratio falls to 2.7, 2.2, 1.5 and 1.1 respectively. Entity fabrication and phantom actors alone carry the effect.

The encoders retain far more under shift, with binary detection at 0.642 to 0.729 against 0.566, and fabrication remains the worst-hit axis for every one. The size of the asymmetry does not track detector strength: DeBERTa at 512 tokens is the strongest detector under shift on both binary AUC and macro loss, yet its ratio of 2.4 exceeds either ModernBERT’s. Within the encoder family it falls as context grows, 2.4 to 1.7 to 1.4, but context does not explain it either, since TF-IDF sees the whole report untruncated and has the largest ratio of all. We cannot say whether the ratio plateaus above one. For screening at volume, the cheapest encoder is thus also the most robust, DeBERTa processing 2.5 times fewer tokens per report than ModernBERT at 1,280.

Context is not the binding constraint. ModernBERT at 1,280 tokens sees what the judge saw, at 2,048 it sees more, and the two are indistinguishable under shift (binary 0.644 against 0.642), consistent with the labels rather than the detector’s input being the limiting factor, though it does not establish that.

Why detectors fail at all is clearer than why fabrication fails most. Scene fabrication prevalence is 89.6% for Claude and 70.7% for Gemini against 30.9% for GPT, so recognising the source and applying its base rate scores well without assessing any claim. Section 4.4 measures how much of the signal that accounts for. Removing identity features does not help; style is recoverable from the text itself (Appendix B).

Transfer also depends on whether a similar failure profile appears in training: holding out GPT, the only omission-dominant family, drops embedding scene-fabrication AUC to 0.527 (Table 8), while holding out Claude raises it to 0.853.

The held-out-strategy split isolates what is specific to the source LLM rather than to distribution shift in general. Training on single-turn strategies and testing on multi-turn ones costs TF-IDF almost nothing: macro 0.846 to 0.837, binary 0.820 to 0.808 (Appendix A), and the other feature sets behave the same way. A detector that has never seen multi-turn reports handles them; one that has never seen a source LLM does not.

Table 3: How much of a detector’s above-chance signal source identity alone supplies. The base-rate predictor sees no report text: it returns the training-set prevalence of its label for the source LLM that wrote the report. Share is $(\text{base-rate} - 0.5)/(\text{tfidf} - 0.5)$, both on the standard video-disjoint split.

Type	Axis	base rate	tfidf	share
H1 Scene Fabrication	fabrication	0.777	0.882	73%
H5 Entity Fabrication	fabrication	0.773	0.888	70%
H6 Phantom Actors	fabrication	0.662	0.859	45%
H2 Crime Misclassification	distortion	0.595	0.773	35%
H4 Severity Minimization	distortion	0.599	0.822	31%
H3 Crime Missed	omission	0.545	0.854	13%
ANY	binary	0.725	0.820	70%

4.4 SOURCE IDENTITY IS RECOVERABLE, AND CARRIES MOST OF THE SIGNAL (RQ3)

The shortcut can be measured rather than inferred. A TF-IDF classifier predicting which of the three source LLMs wrote a report reaches 1.000 accuracy on held-out videos against a chance rate of 0.333.

What that buys depends sharply on the type (Table 3). A predictor with no access to the report, returning only the source LLM’s training-set prevalence, reaches 0.777 on scene fabrication and 0.773 on entity fabrication against the lexical detector’s 0.882 and 0.888. On the above-chance scale that is 73% and 70% of what the detector achieves. On crime omission it reaches 0.545 against 0.854, or 13%. Detection of the fabrication labels is therefore mostly source recognition and omission is mostly not, which is the mechanism behind the transfer asymmetry.

4.5 WHAT GROUNDING BUYS, AND WHAT IT DOES NOT (RQ4)

Among the linear detectors, reference-conditioned features are the only ones that move binary detection under shift, reaching 0.702 against 0.566 for TF-IDF (Appendix G), and they give that family’s best fabrication axis. The encoders exceed that without grounding at all: what grounding buys the linear family, capacity buys the encoder family, and neither removes the asymmetry of Table 2.

One surface cell is diagnostic: on the held-out-Gemini fold severity minimization falls to 0.443, below chance, so the relationship learned in distribution reverses on an unseen source rather than merely weakening, as a shortcut does when its feature is anti-correlated in the held-out family.

Grounding is, however, worse on omission: 0.577 against TF-IDF’s 0.793. Our grounded features are lexical by construction, so we tested a semantic version: a cross-encoder NLI model scores every report sentence against the reference and, in the reverse direction, every reference sentence against the report, aggregated into report-level entailment and contradiction features. The reverse direction is what should capture a dropped criminal act. It helps omission and does not rescue it, 0.577 to 0.625 under shift, while costing elsewhere: the entailment detector reaches 0.632 binary against grounding’s 0.702 and 0.648 on omission in distribution against TF-IDF’s 0.854. Its degradation ratio, 4.8, is the largest of any detector we ran. Whether a report drops the criminal event is therefore not a lexical-representation problem, and claim-decomposed matching (Min et al., 2023) may fare no better. No detector wins everywhere, and binary detection under shift remains the benchmark’s open problem, at 0.729 for the strongest detector against 0.861 in distribution.

4.6 TWO PROPERTIES OF THE LABELS THEMSELVES (RQ5)

Two properties of the labels limit what they can support. The first is a length bias, exposed by scoring a single feature against both the panel and the human labels on the shared 130 reports (Table 4). Report length predicts the panel’s scene-fabrication label at AUC 0.727 but the human label at only 0.566, a gap of 0.161. Severity minimization diverges more sharply still: for humans, length runs inverse to the label (0.275), while the panel sees no relationship (0.505). Entity fabrication and phantom actors show no such gap (0.784 versus 0.800, 0.662 versus 0.668), so the bias is specific rather than pervasive: it says which labels to distrust, not that all are suspect.

Table 4: Does report length predict the panel label or the human label? AUC of a single feature, report word count, against each label source on the 130-report subset. The human column is restricted to reports where both raters agree (n from Table 1). A large positive gap means the feature tracks the judge rather than the phenomenon. Other features are in Appendix F.

Type	AUC vs. panel	AUC vs. human	Gap
H1 Scene Fabrication	0.727	0.566	+0.161
H4 Severity Minimization	0.505	0.275	+0.230
H3 Crime Missed	0.484	0.390	+0.093
H2 Crime Misclassification	0.621	0.608	+0.013
H5 Entity Fabrication	0.784	0.800	−0.016
H6 Phantom Actors	0.662	0.668	−0.006

The divergence is specific to verbosity, not to disagreement in general. The same comparison for the reference-conditioned features gives negative gaps on almost every type, reaching -0.236 on scene fabrication. Only report length predicts the judge better than it predicts a human reader.

Part of the effect is mechanical. Judge inputs cap at 3,000 characters, so above the cap length is unobservable: the truncation flag alone predicts the panel’s scene-fabrication label at 0.686, and within truncated reports length falls to 0.537. It does not dissolve there. On the 45 untruncated reports the panel reaches 0.669 against the human’s 0.566, consistent with the labelling mechanism of Section 4.1, where granular detail absent from a summary-level reference is scored as fabrication. The bias is real but smaller than the pooled 0.727 suggests.

The second property is structural. Excluding self-evaluation has an unavoidable consequence: each source LLM is scored by a different pair of judges. Those pairs differ sharply in mutual agreement. Cohen’s κ on scene fabrication is 0.850, 0.415 and 0.148 for the pairs scoring Claude, GPT and Gemini respectively (Appendix E), so under a conjunctive rule a disagreeing pair yields fewer positives for purely mechanical reasons. The pairwise agreements are published with the corpus; what has not been drawn from them is that under no-self-evaluation they attach to source LLMs rather than to judges. That makes them a confound between the source LLM and its judges rather than a reliability statistic, and it follows from the design rather than from this corpus.

The confound is testable because each judge scores exactly two source LLMs, permitting a within-judge comparison that holds the judge fixed (Appendix E). Across 18 judge-by-type-by-pair comparisons the ordering given by the aggregated labels is reproduced in 16, and at the axis level in all six: every judge that sees GPT calls it omission-dominant, and every judge that sees Claude or Gemini calls them fabrication-dominant.

Both failures fall on the GPT judge and on fabrication types, where it flags nearly everything it sees (92.0% and 96.1% on scene fabrication) and cannot discriminate. The Claude-versus-Gemini fabrication ordering rests on a 4 to 7 point within-judge margin and is judge-dependent; the two orderings involving GPT, at 29 to 41 points and reproduced by every judge, are robust.

5 LIMITATIONS

Three source LLMs. Leave-one-source-out has three folds, thin evidence for a claim about model families in general.

Fabrication labels, not fabrication. Our raters agree least on these types (macro $\kappa = 0.28$) and the panel’s label tracks report length where theirs does not, so a property of fabrication and a property of how it is labelled here cannot be separated. The asymmetry does survive dropping the weakest type, in all four detector families.

One domain. The magnitudes should not be assumed to carry to summarisation, captioning or dialogue; settling that needs a replication on a second corpus crossed over generators.

6 CONCLUSION

Hallucination detectors are evaluated almost exclusively on text from the source LLMs they were trained on, and that evaluation cannot distinguish detecting the phenomenon from recognising the source LLM. In forensic video reporting the source LLM is trivially recoverable, a base-rate predictor that reads no text supplies most of the apparent signal on the fabrication labels, and withholding a source LLM costs those labels several times what it costs omission and distortion, consistently across four detector families. The labels themselves carry a related weakness, tracking report length where human annotators do not, and the no-self-evaluation constraint that makes the panel credible also gives each source LLM its own pair of judges. A held-out-source condition is what separates detection from source recognition, and per-judge verdicts are what make the labels auditable.

ETHICS STATEMENT

The corpus derives from a publicly available surveillance benchmark of de-identified footage; annotators worked from text only. We release the resource for auditing and detector development. It is not suitable for automated legal or investigative decision-making: a system whose error rate depends on the source LLM, the prompting strategy and the event type does not meet the standard evidentiary use would require.

AI USE STATEMENT

An AI assistant was used to draft and revise sections of this paper from the authors' direction.

REPRODUCIBILITY STATEMENT

A single entry point rebuilds every label from the released per-judge verdicts and regenerates every table and figure, verifying the published corpus statistics on each run so that divergence is caught immediately. All splits are seeded. Section 3.3 lists the released artifacts and gives the anonymous repository link.

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APPENDIX

The following material supplements the main text and is referenced from it. Tables are numbered continuously with the main text.

A HELD-OUT-STRATEGY RESULTS

Table 5: Held-out-strategy split: train on the four single-turn strategies, test on the four multi-turn ones. Compare the macro column against the standard split (Table 6) and against leave-one-source-out (Table 2).

Features	H1	H2	H3	H4	H5	H6	ANY	MACRO	MACRO-HQ
surface	0.809	0.601	0.547	0.500	0.800	0.736	0.732	0.665	0.637
surface-noid	0.794	0.593	0.555	0.534	0.790	0.719	0.703	0.664	0.638
tfidf	0.868	0.753	0.859	0.832	0.873	0.839	0.808	0.837	0.831
embed	0.745	0.681	0.617	0.730	0.632	0.624	0.633	0.671	0.657
grounded	0.837	0.690	0.602	0.358	0.821	0.769	0.746	0.679	0.648
DeBERTa-v3, 512	0.842	0.816	0.871	0.879	0.870	0.876	0.810	0.859	0.862
ModernBERT, 1,280	0.826	0.777	0.844	0.863	0.857	0.873	0.787	0.840	0.843
ModernBERT, 2,048	0.828	0.786	0.853	0.873	0.854	0.871	0.785	0.844	0.847

B SAME-SOURCE BASELINES AND THE IDENTITY ABLATION

Table 6: Baseline AUC by split for the surface feature families. The held-out-source rows are the held-out-Gemini fold, not the three-fold mean; Table 7 gives the per-fold values and means for TF-IDF. Removing the source and strategy identity indicators (surface-noid) does not remove the shortcut, because style is recoverable from the text itself: scene fabrication still reaches 0.811 under the standard split and still collapses to 0.603 on this fold.

Features	Split	H1	H2	H3	H4	H5	H6	ANY	MACRO-HQ
surface	random	0.858	0.690	0.646	0.653	0.865	0.772	0.765	0.725
	held-out strategy	0.809	0.601	0.547	0.500	0.800	0.736	0.732	0.637
	held-out src (Gemini)	0.628	0.646	0.506	0.443	0.559	0.564	0.583	0.544
surface-noid	random	0.811	0.667	0.576	0.636	0.830	0.738	0.720	0.689
	held-out strategy	0.794	0.593	0.555	0.534	0.790	0.719	0.703	0.638
	held-out src (Gemini)	0.603	0.614	0.470	0.538	0.567	0.632	0.582	0.564
tfidf	random	0.882	0.773	0.854	0.822	0.888	0.859	0.820	0.839
	held-out strategy	0.868	0.753	0.859	0.832	0.873	0.839	0.808	0.831
	held-out src (Gemini)	0.630	0.727	0.795	0.715	0.621	0.720	0.613	0.716

C PER-FOLD LEAVE-ONE-SOURCE-OUT

Table 7: TF-IDF features, per fold, with bootstrap 95% confidence intervals (1,000 resamples). The crime-omission interval overlaps neither the scene- nor the entity-fabrication interval in any fold, which is the separation cited in Section 4.3.

Held-out source LLM	H1	H2	H3	H4	H5	H6	ANY
Claude	0.596 [.581, .611]	0.718 [.704, .732]	0.787 [.776, .799]	0.788 [.774, .801]	0.671 [.658, .686]	0.778 [.765, .788]	0.539 [.502, .578]
GPT	0.702 [.689, .714]	0.735 [.719, .749]	0.796 [.785, .807]	0.779 [.767, .790]	0.719 [.707, .730]	0.736 [.713, .760]	0.548 [.531, .566]
Gemini	0.630 [.613, .644]	0.727 [.715, .741]	0.795 [.785, .806]	0.715 [.702, .729]	0.621 [.604, .640]	0.720 [.708, .732]	0.613 [.585, .636]
Mean (held-out)	0.643	0.727	0.793	0.761	0.671	0.744	0.566
Random split	0.882	0.773	0.854	0.822	0.888	0.859	0.820
Degradation	-0.239	-0.046	-0.061	-0.061	-0.217	-0.115	-0.254

Table 8: AUC on the held-out source LLM, per fold, for every feature set. Holding out GPT, the only omission-dominant family, drops embedding scene-fabrication AUC to 0.527 while raising crime-omission AUC to 0.824; holding out Claude, leaving the similarly fabrication-dominant Gemini, raises scene fabrication to 0.853.

Features	Held out	H1	H2	H3	H4	H5	H6	ANY
embed	Claude	0.853	0.659	0.689	0.674	0.862	0.661	0.670
	GPT	0.527	0.624	0.824	0.716	0.566	0.673	0.587
	Gemini	0.671	0.627	0.645	0.644	0.655	0.560	0.528
grounded	Claude	0.883	0.612	0.572	0.546	0.840	0.566	0.852
	GPT	0.665	0.596	0.642	0.528	0.639	0.579	0.616
	Gemini	0.649	0.720	0.518	0.523	0.616	0.673	0.638
DeBERTa-v3, 512	Claude	0.934	0.811	0.863	0.847	0.928	0.782	0.816
	GPT	0.708	0.845	0.913	0.864	0.746	0.748	0.665
	Gemini	0.630	0.845	0.889	0.888	0.692	0.818	0.707
ModernBERT, 1,280	Claude	0.885	0.656	0.785	0.731	0.894	0.756	0.739
	GPT	0.548	0.721	0.833	0.812	0.622	0.669	0.592
	Gemini	0.580	0.692	0.707	0.737	0.631	0.777	0.601
ModernBERT, 2,048	Claude	0.877	0.684	0.803	0.750	0.897	0.792	0.738
	GPT	0.602	0.708	0.807	0.783	0.664	0.721	0.601
	Gemini	0.576	0.744	0.798	0.786	0.644	0.787	0.588

D VIDEO-DISJOINT LEAVE-ONE-SOURCE-OUT

Holding out a source LLM leaves the same videos on both sides of the split, since every video is described by every source LLM. Re-running the three folds with video disjointness additionally enforced (training on 9,003 reports, testing on 1,016) lowers every number and widens the gap: the fabrication axis falls to 0.668 and omission to 0.780, so the omission-minus-fabrication margin grows from 0.107 to 0.112, and binary detection falls to 0.559. The figures in the main text are therefore the conservative ones.

E THE JUDGE-PAIR CONFOUND

Table 9: Judge-pair agreement and within-judge axis rates. The pairs differ by a factor of five on the fabrication types; the within-judge rates hold the judge fixed, and the dominant axis is reproduced by every judge for every source LLM it sees.

(a) Pair agreement, Cohen’s κ					
Source LLM	Judged by	H1	H5	overall	
Claude	GPT + Gemini	0.850	0.845	0.809	
GPT	Claude + Gemini	0.415	0.475	0.580	
Gemini	Claude + GPT	0.148	0.138	0.614	

(b) Within-judge axis rates (%)					
Judge	Source LLM	Fabrication	Omission	Distortion	Dominant
Claude	GPT	28.6	55.7	42.7	omission
Claude	Gemini	66.7	54.3	39.8	fabrication
GPT	Claude	77.8	41.7	36.0	fabrication
GPT	Gemini	79.2	49.4	38.1	fabrication
Gemini	Claude	72.7	44.5	26.8	fabrication
Gemini	GPT	44.1	53.1	33.1	omission

F LABEL-SOURCE DIVERGENCE, ALL FEATURES

Table 10: AUC of each single feature against the panel label and against the human label, on the 130-report subset. Table 4 reports the report-length column. Every reference-conditioned feature diverges in the opposite direction: on grounded quantities the panel does not diverge from the human raters.

Type	novel rate		missing rate		hedging rate	
	panel	human	panel	human	panel	human
H1 Scene Fabrication	0.535	0.546	0.300	0.536	0.551	0.451
H2 Crime Misclassification	0.545	0.600	0.500	0.565	0.471	0.573
H3 Crime Missed	0.618	0.644	0.671	0.692	0.522	0.547
H4 Severity Minimization	0.547	0.661	0.608	0.730	0.616	0.636
H5 Entity Fabrication	0.467	0.595	0.254	0.342	0.464	0.577
H6 Phantom Actors	0.567	0.607	0.408	0.450	0.461	0.450

G FEATURE SETS UNDER LEAVE-ONE-SOURCE-OUT

Table 11: Feature sets and encoders under leave-one-source-out, averaged over the three folds. Axis scores are the mean per-type AUC within the axis. Among the linear detectors, reference-conditioned features are the only ones that recover binary detection; the fine-tuned encoders exceed them without grounding.

Features	Fabrication	Omission	Distortion	MACRO	ANY
surface	0.584	0.506	0.545	0.558	0.583
nli-grounded	0.599	0.625	0.526	0.579	0.632
embed	0.670	0.719	0.657	0.674	0.595
tfidf	0.686	0.793	0.744	0.723	0.566
grounded	0.679	0.577	0.587	0.631	0.702
grounded+surface	0.691	0.607	0.599	0.646	0.652
DeBERTa-v3, 512	0.776	0.888	0.850	0.820	0.729
ModernBERT, 1,280	0.707	0.775	0.725	0.724	0.644
ModernBERT, 2,048	0.729	0.803	0.742	0.746	0.642

H RUBRICS AND FULL CORPUS STATISTICS

The verbatim judge rubric, the human-rater instructions, the per-cell hallucination rates for all 24 source-LLM-by-strategy combinations, and the datasheet are in the released artifact, which also regenerates every table above from the raw per-judge verdicts.